# AI-Driven Reconstruction of Life Cycle from the Household Registers of the Joseon Dynasty

Seongmin Mun,
Assistant Professor of English Language and Literature,
Kyungpook National University, Republic of Korea

Donggue Lee,
Assistant Research Professor of Humanities Research Institute,
Ajou Universty, Republic of Korea

Hyoji Ha,
Assistant Research Professor of Humanities Research Institute,
Ajou Universty, Republic of Korea

**Abstract**

This study applies an AI-based individual linking framework to the digitized Daegu-bu Household Registers (Hojeokdaejang) of the Joseon Dynasty (1681–1876) in order to construct longitudinal life cycle datasets for non-noble classes, such as commoners, merchants, and soldiers. The Daegu-bu registers contain approximately 1.76 million individual-level records across 72 variables, covering 17 administrative districts that reflect a socially and occupationally diverse provincial capital. However, the digitized dataset is organized strictly on a year-by-year basis, making it difficult to trace the same individual over multiple years due to name variations, homonyms, and changes in family relationships. To address these issues, we transform the original row-based records into wide-format household data, score 19 family-related variables according to their temporal stability, and vectorize textual data using the Distiluse multilingual BERT embedding model. Candidate matches are identified via cosine similarity (vector-based) and Tanimoto similarity (binary-encoded data), with thresholds applied to ensure high precision. The linking process prioritizes intra-year family linkages before extending connections across years to build individual life cycles. Evaluation on 200 individuals (718 records) from Dongsang-myeon (1681–1774) yields an average accuracy of 82.4% (SD: 3.7%). Our approach demonstrates that AI-assisted linkage can overcome the limitations of manual historical analysis, enabling scalable, quantitative, and temporally coherent studies of social mobility and household structures in the Joseon Dynasty, with strong potential for application to other pre-modern population datasets and broader comparative research.

## 1. Background and Summary

The systematic analysis of large-scale historical documents is essential for gaining an in-depth understanding of past social structures and individual life cycles, and has been increasingly emphasized in interdisciplinary research across sociology, history, and digital humanities. In particular, the integration of data science and artificial intelligence (AI) techniques into historical document analysis expands the scope of traditional source interpretation and offers new possibilities for simultaneously examining both micro- and macro-level transformations in past societies. [1] The Joseon Dynasty (1392–1910) was characterized by a rigid class hierarchy—yangban (nobility), sangmin (commoners), and cheonmin (lowborn)—and studying the social structures, mobility patterns, and life cycle trajectories of each class holds significant value in both sociological and historical scholarship. However, most existing research and surviving historical records focus heavily on the nobility class, limiting the extent to which findings can be generalized to Joseon society as a whole. To investigate the characteristics of life cycles among non-noble class groups—such as commoners, merchants, and military personnel—and to understand structural differences across classes, it is necessary to develop new analytical methodologies capable of systematically linking individual records across extended time periods.

Among the extant household registers, the Daegu-bu Household Register (Hojeokdaejang) is considered particularly valuable due to the richness and quality of its data. [2] Daegu was the capital of Gyeongsang-do, one of the eight provinces of the Joseon Dynasty, and governed 17 administrative districts (myeon), encompassing both urban and rural areas. It served as both a political and economic hub, housing government offices, military outposts, and a vibrant merchant class. This geographic and social diversity means that its registers include households from a wide range of classes and occupations—urban dwellers, landowning elites, merchants, military personnel, farmers, and slaves—providing a uniquely comprehensive view of a highly stratified provincial capital. Such characteristics make the Daegu-bu dataset an ideal source for studying the life cycles and social mobility patterns of non-noble class populations.

The digitized Daegu-bu registers span the years 1681 to 1876 and contain approximately 1,758,301 individual-level records, each classified under 72 variables, providing rich demographic, occupational, and kinship information relevant to the study of social structure and life cycle patterns among commoners. [3] However, the current digital dataset is organized strictly on a year-by-year basis, with each year's records maintained independently. As a result, information on a single individual—such as personal name, family relations, and household details—is fragmented across different time points. Manually linking these dispersed entries into continuous life cycle data is prohibitively laborious, requiring precise identification of name variations, disambiguation of individuals with identical names, and incorporation of changes in family relationships, all of which are impractical given the scale of hundreds of thousands of records.

To address these limitations, we apply an AI-based individual linking algorithm. First, the digitized dataset—originally recorded in row-wise format with each individual occupying a separate entry—is transformed into a wide-format household dataset, with each head of male householder (Juho, 主戶) as the anchor and associated family member information stored as variables. We then score 19 family-related variables (e.g., personal name, father's name, spouse's family background) and vectorize them using the Distiluse multilingual BERT[4] embedding model. Similarity between entries is evaluated using cosine similarity (for vectorized data) and Tanimoto similarity (for binary-encoded data), and candidates exceeding predefined thresholds are selected as potential matches. The linking process prioritizes identifying intra-year household connections to ensure temporal continuity, then iteratively carries forward matches across multiple years to build longitudinal life-history profiles of individuals.

Validation on a test set of 200 individuals from Dongsang-myeon who appeared at least twice between 1681 and 1774—covering 718 total records—yielded an average matching accuracy of 82.4% (standard deviation: 3.7%). These results demonstrate that it is feasible to construct reliable life cycle datasets for large-scale non-noble class populations. By operationalizing a high-precision, AI-assisted linking framework that surpasses the limitations of manual historical analysis, this study establishes a scalable methodological foundation for quantitatively and temporally examining the life cycle and household structures of non-elite populations in the Joseon Dynasty. The proposed approach also holds strong potential for application to other regions, periods, and pre-modern population datasets, thereby enabling broader comparative studies of historical social mobility.

## 2. Methods

### 2.1. Data Cleaning

The dataset used in this study is the Daegu-bu household register, recorded over roughly 200 years from 1681 to 1876. It covers 17 administrative regions and includes people from various social classes — from the upper-class yangban to commoners and slaves (nobi). The records were already digitized by previous research and contain 72 different variables, such as personal name, year of birth, family relationships, household status, occupation, and place of residence. In total, the dataset holds about 1,758,301 individual records, organized by household for each survey year.

The original format follows a hierarchical, row-based structure. Each household begins with the head of household (juho), followed by their wife (cheo), children, slaves (nobi), and other individuals, listed in order. While this format is useful for understanding the family structure and hierarchy at a single point in time, it is less suitable for tracking the same individual over a long period. In particular, non-head members have very limited recorded information, making it difficult to confirm if a person appearing in two different registers is the same. Also, when the same person appears in multiple registers, linking them requires combining year, household, and relationship information — something not easily done with the original layout.

To make the data more suitable for analysis, we applied a series of cleaning and restructuring steps. First, we removed inconsistencies in the original records and standardized variable values. For example, names were unified across Hangul and Hanja spellings, and alternative names or courtesy names were merged according to predefined rules so they could be used as reference information when identifying individuals. Relationship labels were also standardized — for example, variations such as "生父" and "父" were made uniform — to ensure consistency.

Second, we identified and resolved duplicate or contradictory records. If the same individual was recorded multiple times in the same survey year, those entries were merged. When different sources had conflicting information, we determined the final value based on the recency, stability of the relationship, and reliability of the source. This step helped ensure data integrity and reduce potential errors in later matching stages.

Finally, we transformed the row-based structure into a column-based (wide) format, as shown in Figure 1. All household members' information was merged into the row of the head of household. This effectively converted the dataset from "long format" to "wide format." After transformation, the dataset was reduced from 1,758,301 people to 326,773 household heads. The number of columns varied depending on household size. This restructuring allowed each row to hold a complete set of chronological and relationship information, providing a solid basis for the person-linking algorithm. [5,6,7,8]


Figure 1. Conversion of the Daegu-bu household register from row-based to wide format.

2.2. Model Creation

After data cleaning and restructuring, we designed an algorithm to automatically detect and connect repeated appearances of the same individual, as shown in Figure 2. The algorithm combines traditional rule-based matching with AI-driven similarity calculations, so it can handle both the specific characteristics of historical records and the challenges of sparse data.

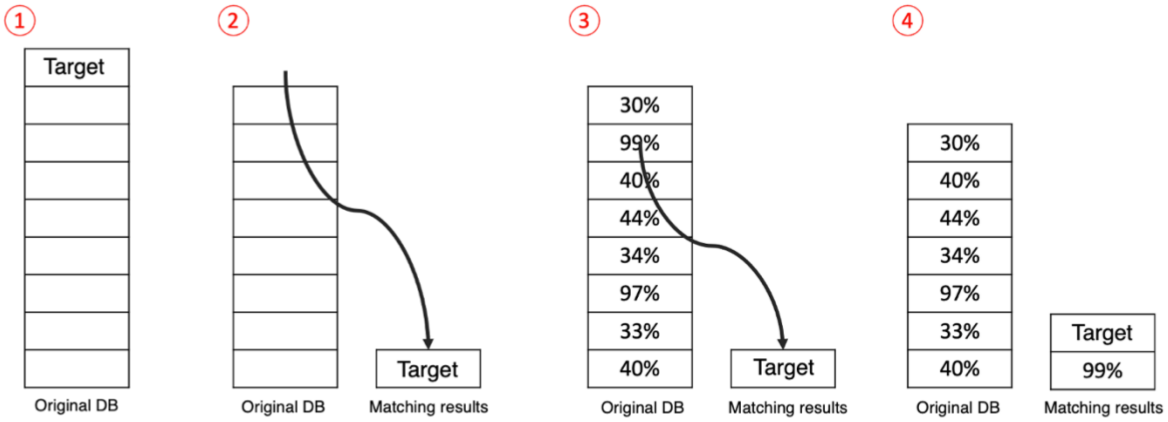


Figure 2. Workflow of the person-linking algorithm combining rule-based and AI methods.

The first step is family relationship–based matching. The algorithm selects the first person in the database as a target and compares them with all others to decide if they are the same individual. Twenty family relationship variables are used for this comparison, including name, father's name, mother's name, and spouse's name. As shown in Table 1, each variable is given a different weight based on prior studies.

| No. | Variable (ENG) | Variable (KOR, CHN) | Weight |
|---|---|---|---|
| 1 | Name_CHN | 姓名 | 5 |
| 2 | Name | 성명 | 5 |
| 3 | Father's Name_CHN | 父名 | 4 |
| 4 | Father's Name | 부명 | 4 |
| 5 | Grandfather's Name_CHN | 祖名 | 3 |
| 6 | Grandfather's Name | 조명 | 3 |
| 7 | Great-Grandfather's Name_CHN | 曾祖名 | 2 |
| 8 | Great-Grandfather's Name | 증조명 | 2 |
| 9 | Maternal Grandfather's Name_CHN | 外祖名 | 2 |
| 10 | Maternal Grandfather's Name | 외조명 | 2 |
| 11 | Wife's Name_CHN | 처_姓名 | 3 |
| 12 | Wife's Name | 처_성명 | 3 |
| 13 | Wife's Father's Name_CHN | 처_父名 | 4 |
| 14 | Wife's Father's Name | 처_부명 | 4 |
| 15 | Wife's Grandfather's Name_CHN | 처_祖名 | 3 |
| 16 | Wife's Grandfather's Name | 처_조명 | 3 |
| 17 | Wife's Great-Grandfather's Name_CHN | 처_曾祖名 | 2 |
| 18 | Wife's Great-Grandfather's Name | 처_증조명 | 2 |
| 19 | Wife's Maternal Grandfather's Name_CHN | 처_外祖名 | 2 |
| 20 | Wife's Maternal Grandfather's Name | 처_외조명 | 2 |

Table 1. Family relationship variables and assigned weights for matching.

For example, parental and spousal information, which tends to be stable over time, is given higher weights, while occupation or residence, which may change more often, gets lower weights. Each matching variable adds to a total score, with a maximum of 60 points. If the score reaches 18 or more, the two records are considered the same person. Once matched, a person is removed from the dataset to prevent duplicate linking.

The second step is AI-based supplementary matching. If family relationship information alone cannot confirm a match, the algorithm calculates similarity in vector space using two methods. First, BERT-based embeddings: we use the Distiluse multilingual model to convert each person's information into a vector and calculate cosine similarity.[9] This method can detect matches even with small spelling differences or partial matches. Second, One-Hot encoding: we convert the presence or absence of attributes into binary values (1 or 0) and then compute the Tanimoto similarity.[10] This method works well with sparse data and relies on exact matches for a more conservative comparison.

If either similarity score passes a set threshold, the records are considered a match. If neither the family-based nor the AI-based method finds a match, the person is registered as a new unique individual. This process is repeated across the entire dataset, ultimately creating a timeline of life histories for each individual.[11]

## 3. Results

To validate the performance of the life-history linking algorithm, we compared its results against a ground truth dataset created by experts through manual matching. The validation dataset was drawn from Daegu-bu's Dongsang-myeon region, covering the years 1681 to 1774, and included individuals who appeared at least twice. The set contained 200 confirmed individuals whose identities were established through historical document review and cross-referencing. This ensured that our benchmark had minimal uncertainty.

The 200 individuals' life records totaled 718 rows, with each row representing a record from a specific year's register. If the same person appeared in different years or households, they were listed separately. The expert-linked dataset served as the gold standard for evaluating algorithm accuracy.

The evaluation procedure worked as follows. First, we randomly shuffled the dataset rows so the algorithm could not rely on chronological clues. This forced it to identify matches purely from relationships and attribute similarities. Second, we ran the algorithm on the shuffled dataset, starting with family-based matching (score $\geq$ 18) and then applying AI-based matching (BERT cosine similarity and One-Hot Tanimoto similarity) for remaining unmatched cases. Third, we compared the algorithm's matches to the expert matches.

Accuracy was calculated as the proportion of correctly predicted matches out of all true matches. To ensure reliability, the entire process was repeated 10 times with different shuffles, and we recorded both the average accuracy and the standard deviation. As shown in Figure 3, the algorithm achieved an average accuracy of 82.4%, with a standard deviation of 3.7%, indicating stable performance across runs.

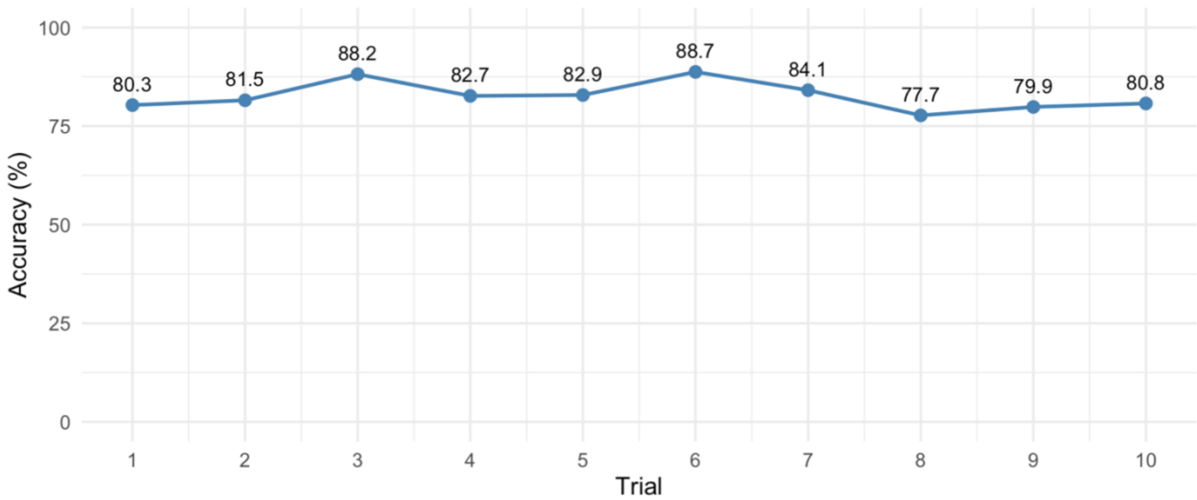


Figure 3. Algorithm accuracy across 10 trials (mean: 82.4%, SD: 3.7%).

Interestingly, the algorithm's matching strategy differed from the experts'. As shown in Figure 4, experts tended to examine records chronologically, linking individuals over time. The algorithm, however, prioritized finding family relationship matches within the same year before moving to the next year. This design choice improved match confidence when relationship data was strong in a given year.

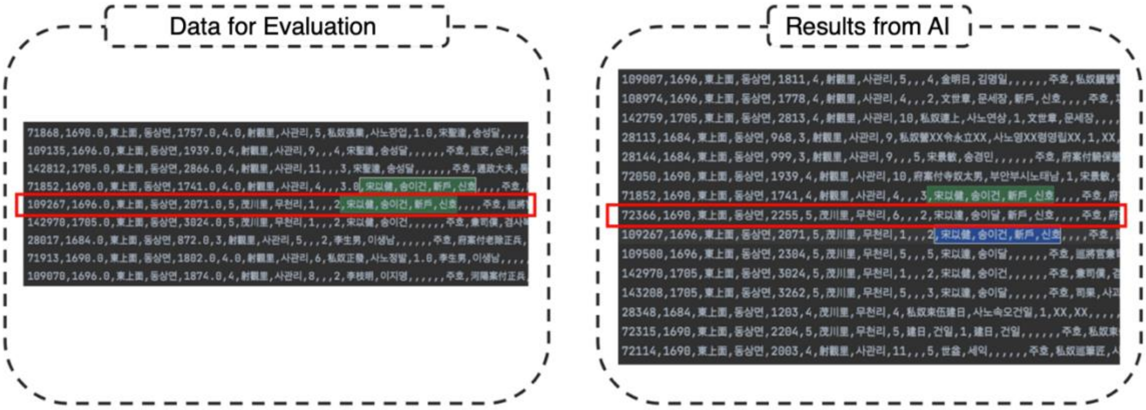


Figure 4. Comparison of matching strategies between expert review and the algorithm.

4. Conclusion

This study demonstrates the feasibility and effectiveness of applying an AI-based individual linking algorithm to large-scale digitized historical records in order to reconstruct longitudinal life cycle data. Focusing on the non-noble class population of the Daegu-bu Household Registers (1681–1876), we developed a hybrid matching process that integrates family-relationship scoring with vector-based similarity measures. The method achieved an average linkage accuracy of 82.4%, indicating its robustness in handling challenges such as name variations, homonyms, and incomplete family data.

In future work, we plan to extend the current framework beyond individual life cycle linkage to develop algorithms capable of connecting household relationships, including father–son and sibling ties. This advancement will enable the reconstruction of inter-generational household networks and facilitate the analysis of social mobility patterns that span multiple generations. Such an extension offers strong potential for application to other regions, historical periods, and pre-modern population datasets, thereby broadening the scope of comparative research in historical demography and digital humanities.

References

[1] Zhang, Y. (2023). Use of artificial intelligence (AI) in historical records transcription: Opportunities, challenges, and future directions. McGill University (Canada).

[2] Daedong Institute for Korean Studies, DataBase, https://ddmh.skku.edu/ddmh/db/intro.do.

[3] The Hoguchongsu (戶口總數), a comprehensive household census conducted in 1789, indicates that Joseon Korea was composed of eight provinces and 334 administrative units known as Gun-

Hyeon (郡縣). The statistical data for Daegu-bu originate from the digitization of its household registers, completed in 2018 by the Daedong Institute for Korean Studies at Sungkyunkwan University, and can be accessed through the following site. https://ddmh.skku.edu/ddmh/db/intro.do

[4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805, 2018.

[5] Lee, J. Z., & Campbell, C. D. (2010). China multi-generational panel dataset, Liaoning (CMGPD-LN), 1749-1909. (No Title).

[6] Wang, H., Chen, S., Dong, H., Noellert, M., Campbell, C., & Lee, J. Z. (2013). China multi-generational panel dataset, Shuangcheng (CMGPD-SC) 1866–1913. User guide. Ann Arbor, MI: Inter-university Consortium for Political and Social Research.

[7] Dong, H., Campbell, C., Kurosu, S., & Lee, J. Z. (2015). Household context and individual departure: The case of escape in three 'unfree'East Asian populations, 1700–1900. Chinese journal of sociology, 1(4), 515-539.

[8] Dong, H., Campbell, C., Kurosu, S., Yang, W., & Lee, J. Z. (2015). New sources for comparative social science: Historical population panel data from East Asia. Demography, 52(3), 1061-1088.

[9] Reimers, N., & Gurevych, I. (2019). Sentence-bert: Sentence embeddings using siamese bert-networks. arXiv preprint arXiv:1908.10084.

[10] Chung, N. C., Miasojedow, B., Startek, M., & Gambin, A. (2019). Jaccard/Tanimoto similarity test and estimation methods for biological presence-absence data. BMC bioinformatics, 20(Suppl 15), 644.

[11] Price, J., Buckles, K., Van Leeuwen, J., & Riley, I. (2021). Combining family history and machine learning to link historical records: The Census Tree data set. Explorations in Economic History, 80, 101391.